

Inspector Assignment

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Mode: Full Pipeline

1. Problem Formulation

Problem Type:	Find
Domain:	Graph Theory / Combinatorics
Named Problem:	Assignment Problem

Universe

- Assignments: $\{ \text{Alice, Bob, Carol, Dave, Eve, Frank} \}$

Variables

Symbol	Meaning	Type / Range
x_{ij}	assignment decision	$\{0, 1\}$
n	number of binary variables = 60	$\mathbb{Z}^+$

Structure

Assign 6 food safety inspectors to 10 facilities (5 types x 2 each) to maximize total expertise score. Each inspector handles at most 3 facilities; each facility needs exactly 1 inspector.

Real-World Mapping

Real-World Concept	Mathematical Object
assignments	x : solution vector (assignments)
total score/cost	objective function value

Constraints

(1) $\sum_j x_{ij} \leq 3$

capacity limit

(2)

$$\sum_j x_{ij} = 1$$

exact assignment

(3)

$$\sum_i x_{ij} = 1$$

coverage requirement

(4)

$$x_{ij} \in \{0, 1\} \quad \forall i, j$$

binary decision variables

Objective

$$\max \sum_{i,j} c_{ij} x_{ij}$$

Approach

Suggested approach: ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: ILP

2. Solution

Answer:	6-element assignments: Alice -> [F1, F2], Bob -> [F3, F4], Carol -> [F5, F6]
Objective Value:	90.0
Optimal:	Yes
Feasible:	Yes
Algorithm:	ILP (PuLP/CBC, Branch & Bound)
Time:	0.0078s
Certificate:	LP relaxation = ILP solution (gap = 0%). Global optimum proven.

Solution Details

Assignments:

Alice -> F1, F2
Bob -> F3, F4
Carol -> F5, F6
Dave -> F7, F8
Eve -> F9, F10
Frank ->

Objective value: $z^* = 90.0$

LP relaxation: 90.0000

Integrality gap: 0.00%

Solution efficiency: 100.0%

Method: Integer Linear Programming via PuLP/CBC.

- 60 binary variables (6 inspectors x 10 facilities)
- 16 constraints (10 facility-coverage + 6 inspector-capacity)
- The LP relaxation is integral (totally unimodular constraint matrix),
so LP optimal = ILP optimal.

Verification

✓ Feasibility	All constraints satisfied
✓ Optimality	LP relaxation = 90.00, gap = 0.00%
✓ Efficiency	100.0% of theoretical maximum

3. Interpretation

The Question

Assign 6 food safety inspectors to 10 facilities (5 types x 2 each) to maximize total expertise score.

The Answer

The optimal solution has objective value $z^* = 90.0$.

What This Means

- Optimal score: 90/90 (100% efficiency)
- Each facility assigned to its top expert
- 0% optimality gap

Integer Linear Programming via PuLP/CBC.

- 60 binary variables (6 inspectors x 10 facilities)
- 16 constraints (10 facility-coverage + 6 inspector-capacity)
- The LP relaxation is integral (totally unimodular constraint matrix), so LP optimal = ILP optimal.

Sensitivity Analysis

Parameter	Current	Change	New Obj.	Class
Alice	present	removed	88.0	sensitive
Bob	present	removed	85.0	critical
Carol	present	removed	85.0	critical
Dave	present	removed	86.0	sensitive
Eve	present	removed	87.0	sensitive
Frank	present	removed	90.0	robust

Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

- Solution